HPXN302-1 Praxis N3B SA2 January - June

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Name of your Support Centre: BELLVILLE

Code:

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// ------------------------------------------------------------------------------------------

// Automated Licence Compliance Management System (ALCMS)

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// 1. DATA INTEGRATION LAYER

// Simulated databases


const sabcLicenceDB = [

  { id: "LIC001", idNumber: "8501015800083", name: "John Stemmet", paid: true, expiry: "2026-12-31" },

  { id: "LIC002", idNumber: "9003025800085", name: "Thandi Nkosi", paid: false, expiry: "2025-06-30" },

  { id: "LIC003", idNumber: "7812125800087", name: "Miriam Dlamini", paid: true, expiry: "2026-08-15" },

];


const retailTVSalesDB = [

  { idNumber: "8501015800083", name: "John Stemmet", tvPurchaseDate: "2024-03-15", retailer: "Game" },

  { idNumber: "9003025800085", name: "Thandi Nkosi", tvPurchaseDate: "2023-11-20", retailer: "Makro" },

  { idNumber: "7812125800087", name: "Miriam Dlamini", tvPurchaseDate: "2025-01-10", retailer: "HiFi Corp" },

  { idNumber: "[CREDIT_DEBIT_CARD_NUMBER]", name: "Lerato Molefe", tvPurchaseDate: "2025-05-22", retailer: "Incredible Connection" },

  { idNumber: "9505055800081", name: "Michael Tau", tvPurchaseDate: "2024-08-30", retailer: "Game" },

  { idNumber: "7001015800082", name: "Dept of Education", tvPurchaseDate: "2024-02-14", retailer: "Makro" },

];
```

```
const homeAffairsDB = [
  { idNumber: "8501015800083", name: "John Stemmet", address: "12 Main Rd, Soweto" },
  { idNumber: "9003025800085", name: "Thandi Nkosi", address: "45 Church St, Pretoria" },
  { idNumber: "7812125800087", name: " Miriam Dlamini", address: "78 Long St, Cape Town" },
  { idNumber: "[CREDIT_DEBIT_CARD_NUMBER]", name: "Lerato Molefe", address: "23 Vilakazi St, Johannesburg" },
  { idNumber: "9505055800081", name: "Michael Tau", address: "9 Nelson Mandela Dr, Durban" },
  { idNumber: "7001015800082", name: "Dept of Education", address: "222 Struben St, Pretoria" },
];

// 2. GAP IDENTIFICATION ENGINE
function identifyNonCompliantHouseholds() {
  const nonCompliant = [];

  retailTVSalesDB.forEach((buyer) => {
    const licence = sabcLicenceDB.find((lic) => lic.idNumber === buyer.idNumber);

    if (!licence) {
      // TV owner with NO licence at all
      nonCompliant.push({
        idNumber: buyer.idNumber,
        name: buyer.name,
        status: "NO_LICENCE",
        tvPurchaseDate: buyer.tvPurchaseDate,
        retailer: buyer.retailer,
      });
    } else if (!licence.paid) {
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      // Has licence but hasn't paid
      nonCompliant.push({
        idNumber: buyer.idNumber,
        name: buyer.name,
        status: "UNPAID",
        licenceId: licence.id,
        expiry: licence.expiry,
      });
    } else if (new Date(licence.expiry) < new Date()) {
      // Licence expired
      nonCompliant.push({
        idNumber: buyer.idNumber,
        name: buyer.name,
        status: "EXPIRED",
        licenceId: licence.id,
        expiry: licence.expiry,
      });
    }
  });

  return nonCompliant;
}
```

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// --- 3. AUTOMATED COMMUNICATION MODULE ---
function sendNotifications(nonCompliantList) {
  const notifications = [];

  nonCompliantList.forEach((person) => {
    const contact = homeAffairsDB.find((h) => h.idNumber === person.idNumber);
    let message = "";

    switch (person.status) {
      case "NO_LICENCE":
        message = `Dear ${person.name}, our records indicate you purchased a TV on ${person.tvPurchaseDate} from ${person.retailer} but have no valid SABC TV licence. Please register and pay at https://tvlicence.sabc.co.za within 30 days to avoid penalties.`;
        break;
      case "UNPAID":
        message = `Dear ${person.name}, your TV licence (${person.licenceId}) remains unpaid. Please make payment at https://tvlicence.sabc.co.za to avoid legal action.`;
        break;
      case "EXPIRED":
        message = `Dear ${person.name}, your TV licence (${person.licenceId}) expired on ${person.expiry}. Please renew at https://tvlicence.sabc.co.za.`;
        break;
    }

    notifications.push({
      recipient: person.name,
      address: contact ? contact.address : "Unknown",
      channel: "SMS + Email",
      message: message,
      sentDate: new Date().toISOString().split("T")[0],
```

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    });
  });

  return notifications;
}

// --- 4. SELF-SERVICE PAYMENT PORTAL (Simulation) ---
function processPayment(idNumber, amount) {
  const licence = sabcLicenceDB.find((lic) => lic.idNumber === idNumber);

  if (licence) {
    licence.paid = true;
    licence.expiry = new Date(new Date().setFullYear(new Date().getFullYear() + 1))
      .toISOString()
      .split("T")[0];
    return { success: true, message: `Payment of R${amount} received. Licence renewed until
${licence.expiry}.` };
  } else {
    // Create new licence
    const newLicence = {
      id: `LIC${String(sabcLicenceDB.length + 1).padStart(3, "0")}`,
      idNumber: idNumber,
      name: homeAffairsDB.find((h) => h.idNumber === idNumber)?.name || "Unknown",
      paid: true,
      expiry: new Date(new Date().setFullYear(new Date().getFullYear() + 1))
        .toISOString()
        .split("T")[0],
    };
    sabcLicenceDB.push(newLicence);
```

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    return { success: true, message: `New licence ${newLicence.id} created. Valid until
${newLicence.expiry}.` };
  }
}

// --- 5. COMPLIANCE DASHBOARD ---
function generateComplianceDashboard() {
  const totalTVOwners = retailTVSalesDB.length;
  const compliantOwners = retailTVSalesDB.filter((buyer) => {
    const licence = sabcLicenceDB.find((lic) => lic.idNumber === buyer.idNumber);
    return licence && licence.paid && new Date(licence.expiry) >= new Date();
  }).length;

  const complianceRate = ((compliantOwners / totalTVOwners) * 100).toFixed(1);
  const nonCompliant = identifyNonCompliantHouseholds();
  const estimatedRevenueLoss = nonCompliant.length * [CREDIT_DEBIT_CARD_CVV]; //
R[CREDIT_DEBIT_CARD_CVV] annual TV licence fee

  return {
    totalTVOwners,
    compliantOwners,
    nonCompliantOwners: nonCompliant.length,
    complianceRate: `${complianceRate}%`,
    estimatedAnnualRevenueLoss: `R${estimatedRevenueLoss.toLocaleString()}`,
    nonCompliantBreakdown: {
      noLicence: nonCompliant.filter((p) => p.status === "NO_LICENCE").length,
      unpaid: nonCompliant.filter((p) => p.status === "UNPAID").length,
      expired: nonCompliant.filter((p) => p.status === "EXPIRED").length,
    },
  };
```

```
}
```

```
// ------------------------------------------------------------------------------------------
// RUNNING THE SYSTEM
// ------------------------------------------------------------------------------------------

console.log("=== SABC ALCMS - System Execution ===
");

// Step 1: Identify non-compliant households
console.log("--- STEP 1: Gap Identification ---");
const nonCompliant = identifyNonCompliantHouseholds();
console.log(`Found ${nonCompliant.length} non-compliant TV owner(s):`);
nonCompliant.forEach((p) => console.log(`  • ${p.name} - Status: ${p.status}`));

// Step 2: Send automated notifications
console.log("
--- STEP 2: Automated Notifications ---");
const notifications = sendNotifications(nonCompliant);
notifications.forEach((n) => console.log(`  → Sent to ${n.recipient}: ${n.message}
`));

// Step 3: Simulate a payment
console.log("--- STEP 3: Payment Processing ---");
const payment = processPayment("[CREDIT_DEBIT_CARD_NUMBER]",
[CREDIT_DEBIT_CARD_CVV]);
console.log(`  ${payment.message}`);

// Step 4: Generate dashboard
console.log("
--- STEP 4: Compliance Dashboard ---");
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const dashboard = generateComplianceDashboard();
console.log(`  Total TV Owners:       ${dashboard.totalTVOwners}`);
console.log(`  Compliant:             ${dashboard.compliantOwners}`);
console.log(`  Non-Compliant:         ${dashboard.nonCompliantOwners}`);
console.log(`  Compliance Rate:       ${dashboard.complianceRate}`);
console.log(`  Est. Revenue Loss:     ${dashboard.estimatedAnnualRevenueLoss}`);
console.log(`  Breakdown:`);
console.log(`    - No Licence:        ${dashboard.nonCompliantBreakdown.noLicence}`);
console.log(`    - Unpaid:            ${dashboard.nonCompliantBreakdown.unpaid}`);
console.log(`    - Expired:           ${dashboard.nonCompliantBreakdown.expired}`);
```